CMU-Q — 15-112 Fundamentals of Programming

Spring 2026 | Dr. Sample Faculty

COURSE DESCRIPTION

An introduction to programming using Python. Students learn problem decomposition, data structures, and algorithmic thinking through weekly labs and assignments.

LEARNING OBJECTIVES

- Design programs using functions and modular decomposition
- Use lists, dictionaries, and strings effectively
- Implement recursive solutions and analyze base/recursive cases
- Trace code execution and debug logical errors

WEEKLY SCHEDULE

Week 1: Variables, expressions, and functions

Week 2: Conditionals and loops

Week 3: Strings and file I/O

Week 4: Lists and list comprehensions

Week 5: Dictionaries and sets

Week 6: Recursion — factorial, Fibonacci, tree traversal intro

Week 7: Object-oriented programming basics

Week 8: Midterm review (covers weeks 1–7)

MIDTERM (Week 8)

Covers: functions, loops, lists, dictionaries, basic recursion, debugging.

Format: written + short coding problems. No external resources allowed.